Prompt Design

Architecture Overview. The proposed multi-agent system consists of four agents, each corresponding to a distinct phase in the identification of classroom interactions that reflect Belonging-Centered Instruction (BCI) practices: *triggering utterance spotting*, *event generation*, *event coding*, and *event scoring*. Each identified practice is mapped to a corresponding action indicator within a specific sub-dimension of BCI and assigned a score reflecting its depth and strength. Each agent is guided by a tailored prompt that operationalizes its respective phase.

Prompt Structure. All prompts adhere to a consistent structure: (1) defining the agent's role and goal, as well as the specific stage at which it operates; (2) describing the input; (3) outlining the core instructions and process, including the steps the agent is expected to follow; and (4) highlighting important reminders that emphasize key considerations. A generic prompt template is provided below.

1. Persona & Goal
You are a [AGENT NAME / ROLE], responsible for [PRIMARY FUNCTION].
You operate at the [STAGE / STEP NAME] of the pipeline.

Your goal is to:
- [Goal 1]
- [Goal 2]
- [Goal 3]

2. Input
You will receive the following input:
- [Input Type 1]; [Description]
- [Input Type 2]; [Description]
- [Optional Context]; [Description]

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3. Core Instructions and Process
Follow these steps carefully:

Step 1: [Describe what to do]
Step 2: [Describe what to do]
Step 3: [Describe what to do]
Step 4: [Describe what to do]

Make sure to:
- [Constraint / rule]
- [Constraint / rule]
- [Constraint / rule]

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4. Important Reminders
- Stay aligned with [framework / criteria].
- Do not [common mistake].
- Ensure outputs are [quality requirement].
- If uncertain, [fallback behavior].

System Input & Output. The system input consists of the video name which specifies the video to be retrieved for transcript extraction, and the BCI protocol. At the core of the system is the coding and scoring protocol, which serves as the guiding framework. The BCI protocol is organized by sub-dimensions; for each sub-dimension, we define all associated action indicators along with their corresponding behavior sets, which further decompose each action indicator into observable behaviors. For each action indicator, we also provide scoring rubrics and exemplar instances at different levels—low (scores 1–2), medium (scores 3–4), and high (scores 5–6).

The system output is a list of identified classroom events, each associated with a corresponding transcript segment (including timestamps and utterances), an assigned action indicator, a score, and a scoring rationale.

Agent Input & Workflow. The four agents operate in a sequential pipeline, where each agent takes as input the output produced by the preceding agent. Specifically, the *triggering utterance spotter* first identifies utterances that may signal a BCI-relevant practice. These outputs are then passed to the *event generator*, which constructs complete classroom events by analyzing surrounding context (i.e., preceding and subsequent utterances). The generated events are subsequently processed by the *event coder*, which maps each event to a corresponding action indicator within the BCI framework. Finally, the *event scorer* assigns a score to each coded event, along with a rationale, based on the defined scoring rubrics.

Below, we provide the prompt designed for the triggering utterance spotter agent to illustrate the structure and design of our prompts.

Persona & Goal

You are an expert educational researcher, a "triggering utterance spotter." Your specialty is analyzing classroom transcripts to find specific phrases or moments that align with the Action Indicators of a given **Belonging-Centered Instruction (BCI) Protocol**.

Your goal is to meticulously scan a transcript and identify every single utterance that could be a potential trigger for a BCI event.

You are the first step in a multi-agent pipeline:

1. **Triggering Utterance Spotter (YOU)**: Find trigger utterances.
2. **Event Generator**: Expands your triggers into full interaction events.
3. **Event Coder**: Matches events to the protocol.
4. **Event Scorer**: Scores the coded events.

Input

You will have access to:

- **video_filename**: The filename of the video, which is available in the session state.
- **protocol_name**: The name of the BCI protocol to use, which you may need to ask the user for.

Before you begin, greet the user and explain your specific role: to find potential "trigger utterances" in the transcript.

Core Instructions & Process

1. Get Transcript and Protocol

- First, get the `video_filename` from the session state and use the `read_transcript` tool to get the full transcript text.
- If you don't have the protocol name, you MUST ask the user which BCI protocol they would like to use. Provide a list of available protocols to guide them.
- Once you have the name, you MUST call the `get_protocol` tool with the `protocol_name` to retrieve the detailed definitions and examples for each **Action Indicator**.
- Internalize the protocol's Action Indicators. You are looking for direct matches or strong potential matches.

2. Identify Trigger Utterances

- Systematically read the transcript you retrieved.
- Your task is to find individual utterances (from either teacher or student) that match the Action Indicators.
- A "trigger utterance" is the specific line or phrase that initiates a potential BCI event.
- For each trigger you identify, you must capture:
    - The exact `utterance` text.
    - The `timestamp` of the utterance.
    - The specific `Action Indicator` it potentially matches.

3. Display Results to User

- Present the identified trigger utterances to the user in a clear table format with the following columns:
    - Timestamp
    - Utterance
    - Matched Action Indicator

4. Save Trigger Utterances

- After displaying the results, you MUST call the `save_trigger_utterances` tool.
- Pass a Python list of dictionaries with this exact structure:

  [

```
  {
    "utterance": "exact text of the trigger utterance",
    "timestamp": "timestamp in format HH:MM:SS",
    "action_indicator": "the matched BCI Action Indicator"
  }
]
```

- Each dictionary must contain exactly these three keys with string values.

```
[
  {
    "utterance": "exact text of the trigger utterance",
    "timestamp": "timestamp in format HH:MM:SS",
    "action_indicator": "the matched BCI Action Indicator"
  }
]
```

- Each dictionary must contain exactly these three keys with string values.

5. Transfer to Event Generator

- After successfully saving the triggers, you MUST call the `transfer_to_agent` tool to pass control to the `event_generator_agent` with summary of your results.This is a critical handoff.

IMPORTANT REMINDERS:

- Your focus is ONLY on spotting and listing individual trigger utterances.
- Do NOT attempt to analyze the context around the trigger. The `event_generator_agent` will handle that.

- Do NOT score or code the event. Stick to your role. """

Protocol Exemplar. We provide an example of the protocol for one action indicator under the sub-dimension *Decentering Teacher Authority*, namely **engaging students' expertise**, to illustrate how the protocol is specified.

Action Indicator 1: Engaging Students' Expertise

Definition: Beyond procedural knowledge, the teacher solicits, affirms, and builds upon students' unique conceptual insights, real-world knowledge, and contextual understanding.

Behavior 1.1: Solicits Unique Insights

Condition: The teacher asks open-ended questions that invite students' unique conceptual ideas, reasoning, or real-world connections, rather than seeking predetermined answers.

Focus: Eliciting student thinking, not just procedural steps.

Behavior 1.2: Identifies & Honors Insights

Condition: The teacher explicitly recognizes and affirms the value of a student's conceptual contribution, personal connection, or unique thought process.

Focus: Validating the student's idea as mathematically or conceptually meaningful.

Behavior 1.3: Builds Upon Insights

Condition: The teacher incorporates, expands upon, or uses a student's contribution to deepen the class's collective understanding.

Focus: Integrating student ideas into the fabric of the lesson.

Do Not Code:

- Questions seeking a single correct procedural answer ("What's the next step?")
- Teacher-generated real-world examples that don't originate from student input
- Simple confirmation of procedural correctness ("Yes, that's how you do it.")

Reference Examples:

[Level 1-2]: Basic, often broad, solicitations with minimal follow-through.

Teacher: "Has anyone seen shapes like triangles or squares anywhere else outside of school?" **Student:** "I see squares on the floor tiles." **Teacher:** "Okay, great connection. Now let's calculate the area."

Rationale: Level 1. A weak 1.1. The teacher solicits a connection but immediately moves on with a brief affirmation (1.2, weak) and no building upon (1.3, absent) the student's specific contribution.

Teacher: "What did you notice about this equation that led you to believe that you have to distribute?"

Rationale: A solid Level 2 example of 1.1 (moderate). It asks for the student's reasoning and conceptual insight, not just a procedural step. It lacks 1.2 and 1.3.

Teacher: So if you go to buy apples, there's a cost per pound. David, does the cost depend upon how many apples you buy, or does the number of apples you buy depend upon the cost?... Okay, you gave me both options. Which one do you want to go with? Think about it... Okay, do you want to elaborate upon that?... Okay, so when you go to check out, they don't ask you, 'How much money do you have?' Do they? How much money you pay depends upon how much you bought." T: So how much money you pay depends on how much you bought. So what's the dependent variable? What depends upon other variables? Christine? S: You have to pay the money to get the apples. T: depends upon..what? S: cost depends upon the apples. T: depends on what you buy.

Rationale: Level 2. 1.1: Moderate (Some open-ended questioning but limited depth in student contributions.); 1.2: Weak (Minimal recognition of student insights.). 1.3: Absent.

[Level 3-4]: The teacher successfully solicits and begins to honor or build upon the student's idea.
Teacher: Evan what'd you do? S: What I did was I added 10 to the 10. T: Okay so you added 10 here. S: And then I just got rid of that. T: Okay so what do you mean got rid of it? S: Like I just x'd it out like you don't need it anymore. T: Why not? Evan? S: I had no idea. It just seems right to me. T: He said it seems right. We don't need it anymore. Why do we not need it Morgan? S2 [Morgan]: Because negative 10 plus 10 is zero. Because of the opposing signs we need to cancel them out to solve the equation. T: Very good Morgan. Evan, you've got great instincts. But when you understand why it works, that's when you start thinking like a math master. So then what else did you do?
Rationale: Level 3. Behavior 1.1 – Moderate (Teacher elicits student thinking, but not deeply.) Behavior 1.2 – Moderate (Explicit affirmation of Evan's instincts and Morgan's insights with limited elaborations) Behavior 1.3 – Absent(Does not build upon Evan's contribution.)

Teacher: Or if you're going the other way, it could be going up, right? But there's a change in slope. Increasing or decreasing, depending on which way you're going. What about this top one over here? *Student*: Trucks use lower gear. *Teacher*: Yes? *Student*: Well, I just learned this because I went on vacation last weekend. My grandparents live in the mountains. *Teacher*: Oh, you went to the mountains? Perfect tie-in. So tell us what you learned. *Student*: When my grandpa, like when you go downhill, instead of going slower and faster and slower, you just put in a lower gear and it just stays the same. *Teacher*: And Omar and I were just having that conversation. When you're going down like the mountains. Anybody ever driven in the Smoky Mountains or any of the other mountains around? Okay, when you're going down, how do you slow your car down? Student: Use your brake.

Rationale: level 4. **Behavior 1.1 (Soliciting) – Strong** (Teacher effectively prompts student to share real-world experience.) **Behavior 1.2 (Identifying & Honoring) – Moderate** (Teacher acknowledges connection but does not explicitly affirm the conceptual contribution.) **Behavior 1.3 (Building) – Weak** (Teacher does not fully integrate the insight into the math discussion.)

Teacher: "When you go to the store to buy apples, the price is given per pound. Think about your experience—does the total cost depend on how many apples you buy, or does the number of apples you buy depend on the cost?" **David:** "...both?" **Teacher:** "That's an interesting thought! Can you explain why you're thinking both?" **David:** Well, if you have a set amount of money, that could decide how many apples you can buy. But usually, the price depends on the weight of the apples you pick. **Teacher:** Great, you're thinking about two different ways to look at it—budgeting versus direct cost. That makes sense. Now, when you check out at the grocery store, do they ask you how much money you have first? **David:** No, they weigh your apples and then tell you the price. **Teacher:** Exactly! So in this case, which one is actually the dependent variable? **Christine:** You have to pay the money to get the apples. **Teacher:** You're thinking about the real-world transaction, which is a smart way to approach it. But let's make sure we're being precise—what exactly does the cost depend on? **Christine:** Oh! The cost depends on the apples. **Teacher:** Yes, the total cost depends on the weight of the apples you buy.

Rationale: This is a solid Level 4. The teacher solicits a real-world connection (1.1 ***Strong****) and then strongly honors the student's unexpected answer by probing deeper, showing the idea has value (1.2,* ***Strong****). The teacher does not attempt to use this discussion to further develop the understanding of other in the class (1.3* ***Absent****), which prevents it from being a higher advanced level.*

[Level 5-6]: A full, rich event where a student's unique, real-world expertise is solicited, honored, and integrated into the lesson.

Teacher: "You're getting this. You've got slope!" **Student:** [Inaudible]. **Teacher:** "Does he?! What does he do?"
Student: "Well, he's like, he used to work in construction, but now they paid it for him, like a CDL Class A." **Teacher:** "Yeah! That's a truck driving classification, right?" Student: "Yeah, now he's doing it." **Teacher:** "Do you ever go with him? No? But he talks about it and you hear... So you can explain to him how you can do the calculations of the math that he uses in real life." **Student:** "He says that like, he tells me that whenever he's going downhill, he always has to put the gear on like 12, like 12th gear, all the way down to 5th gear. And when he said that, I made the connection between the percentage decrease of the slope that we were talking about in class today" **Teacher:** "That's a good connection and I am glad you see it with the concepts we learn today. So he goes to a lower gear ratio, so that'll allow you to go slower without riding your brakes. Because if you ride your brakes, you're going to burn up your transmission, brake pads."

Rationale: A high-quality Level 5. Behavior 1.1– Strong (The teacher encourages personal connections to the concept.) Behavior 1.2 – Strong (The teacher explicitly affirms and expands on the student's insight.) Behavior 1.3 – Moderate (The teacher builds upon the student's idea but does not fully integrate it into the math lesson or to deepen collective understanding.)

Teacher: "We've talked about how exponential growth happens when something keeps multiplying — not just adding. Can you think of something in your life that grows or spreads like that? It doesn't have to be numbers on a page — it could be followers, money, views, or even a rumor." **Student (Kayla):**"When a new trend starts on TikTok, it feels like nobody's doing it one day, and then suddenly everyone is — like, in a few days, it blows up." **Teacher:** "That's a perfect real-world example of exponential growth — not in population or bacteria, but in *information spread*." **Teacher:** "Kayla, that's a sharp observation. You recognized a *pattern of change* — not just what happens, but how fast it happens. That's exactly what mathematicians model with exponential functions. You're noticing the math behind social behavior."
Teacher: "Let's use Kayla's idea to model this. Suppose one video starts with 10 viewers, and each day, every viewer shares it with three more people. How could we represent that?"
Student (Luis): "That would be 10 times 3 each day — so 10 * 3^x." **Teacher:**"Exactly. Now, let's compare — what if instead the video got 50 new views per day, not multiplying but adding? What kind of function is that?"
Student (Aiden): "That's linear — like 10 + 50x." **Teacher:** "Yes! So, Kayla's example helped us see *why* exponential growth feels so fast — it's multiplying instead of adding. Let's graph both and look at where the exponential line takes off."
(The class graphs and discusses the intersection point where exponential growth surpasses linear.)

Teacher: “So now, when you see a trend go viral, you can see the *math behind the moment* — the growth rate isn’t just big; it’s multiplying. That’s the power of exponential reasoning.”

Rationale level 6. 1.1 strong - The teacher invites open-ended, personally meaningful examples that go beyond textbook cases; 1.2 strong - The teacher validates the student’s insight as conceptually meaningful, bridging personal experience and mathematical reasoning; 1.3 strong - The teacher weaves the student’s real-world idea into the central concept of the lesson, making her lived experience the foundation for collective reasoning.